

A PROJECT FINAL REPORT

On

E-Commerce Business Management System

Project Type: Full-Stack E-Commerce Web Application

Submitted to:

Primeasia University
For the degree of
Computer Science and Engineering (CSE)

Submitted by:

- | | |
|------------------------------------|------------------------|
| 1. Name: Md. Jobaidul Islam | ID: 242-008-042 |
| 2. Name: Asib Hossen | ID: 242-011-042 |
| 3. Name: Ziaur Rahman Arnab | ID: 242-045-042 |

Under the guidance of:

Halima Mowla
Lecturer at Primeasia University

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Database System Design (Lab)

Course Code:

CSE 232

Department of Computer Science and Engineering (CSE)

Primeasia University
Dhaka, Bangladesh

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Acknowledgement

First of all, we express our sincere gratitude to Almighty Allah for giving us the strength and ability to complete this project successfully.

We would like to express our heartfelt thanks to our course teacher for providing valuable guidance, suggestions, and encouragement throughout the development of this project. Their support helped us understand the practical implementation of database management systems and web application development.

We are also thankful to our friends and classmates who supported us during the project work. Finally, we would like to acknowledge all the online resources and documentation that helped us complete this project successfully.

Abstract

ShajBari Fashion House is a full-stack e-commerce web application developed for the Database Management System (DBMS) course project. The project is designed to provide an online shopping platform for traditional Bangladeshi fashion products such as Sarees and Panjabis.

The system allows users to browse products, search and filter items, add products to a shopping cart, and place orders through a Cash on Delivery system. Registered users can create accounts, log in securely, and view their previous order history from their personal profile dashboard.

The frontend of the application was developed using HTML5, CSS3, and Vanilla JavaScript, while PHP and MySQL were used for backend development and database management. The project implements authentication, session management, AJAX-based communication, and secure database operations using prepared statements.

This project demonstrates the practical implementation of database design, relational database management, user authentication systems, and modern web development concepts.

Table of Contents

1. Introduction	04
2. Objectives of the Project	04
3. Project Overview	04
4. System Analysis	05
5. System Design	06
6. Database Design and ER Diagram	07
7. Technologies Used	10
8. Features of the System	11
9. Security Practices	12
10. User interface and Implementation Details	12
11. Testing and Validation	16
12. Challenges Faced	17
13. Future Improvements	17
14. Conclusion	18
15. References	18

1. Introduction

E-commerce has become one of the most important sectors in modern business systems. Online shopping platforms allow customers to purchase products from anywhere using internet-based systems. In Bangladesh, the demand for online fashion stores is rapidly increasing due to convenience, accessibility, and digital transformation.

The purpose of this project is to develop a complete web-based e-commerce system named “ShajBari Fashion House.” The system focuses on traditional Bangladeshi fashion products such as Sarees and Panjabis. The website provides customers with a user-friendly interface for browsing products, managing carts, and placing orders.

This project also demonstrates practical implementation of Database Management System concepts including relational databases, data storage, relationships, authentication systems, and database-driven applications.

2. Objectives of the Project

The main objectives of this project are:

- To develop a responsive e-commerce website.
- To implement user authentication and session management.
- To create a relational database using MySQL.
- To allow customers to place orders online.
- To maintain order history for registered users.
- To implement secure database communication using prepared statements.
- To gain practical experience in full-stack web development.
- To apply DBMS concepts in a real-world application.

3. Project Overview

Project Name: E-Commerce Business Management System

Website Name: SajBari Fashion House

Project Type: Full-Stack Web Application

Tech Stack:

- HTML5
- CSS3
- JavaScript (Vanilla)
- PHP
- MySQL

Database Name: `project_db`

Main Purpose

The main purpose of this project is to provide an online platform where users can browse fashion products, place orders, and manage their shopping activities.

4. System Analysis

Functional Requirements

The system provides the following functionalities:

User Features

- User registration
- User login/logout
- Product browsing
- Product search and filtering
- Shopping cart system
- Order placement
- Profile management
- Order history tracking

Admin/System Features

- Database management
- Order storage
- Contact message storage
- Session handling

Non-Functional Requirements

- Responsive design
- Secure authentication
- Fast performance
- User-friendly interface
- Data consistency
- Reliable database operations

5. System Design

File Structure

/ (project)

├── index.html

├── login.html

├── register.html

├── profile.html

|

├── css/

| ├── style.css

| ├── login.css

| ├── register.css

| └── profile.css

|

└── php/

├── db.php

├── register.php

├── login.php

├── logout.php

├── check_session.php

├── get_profile.php

├── get_orders.php

├── order.php

└── contact.php

System Workflow

1. User visits the homepage.
2. User browses available products.

3. User adds products to the cart.
4. User places an order.
5. Order information is stored in the database.
6. Registered users can log in and view order history.

Page Connection Flow

index.html —fetch—▶ php/check_session.php

—fetch—▶ php/order.php

—fetch—▶ php/contact.php

login.html —POST—▶ php/login.php

register.html —POST—▶ php/register.php

profile.html —fetch—▶ php/get_profile.php

—fetch—▶ php/get_orders.php

(logout) —▶ php/logout.php

6. Database Design and ER Diagram

Database Name: `project_db`

Tables Used

Table Name	Purpose
users	Stores user account information
orders	Stores order details
order_items	Stores products inside each order

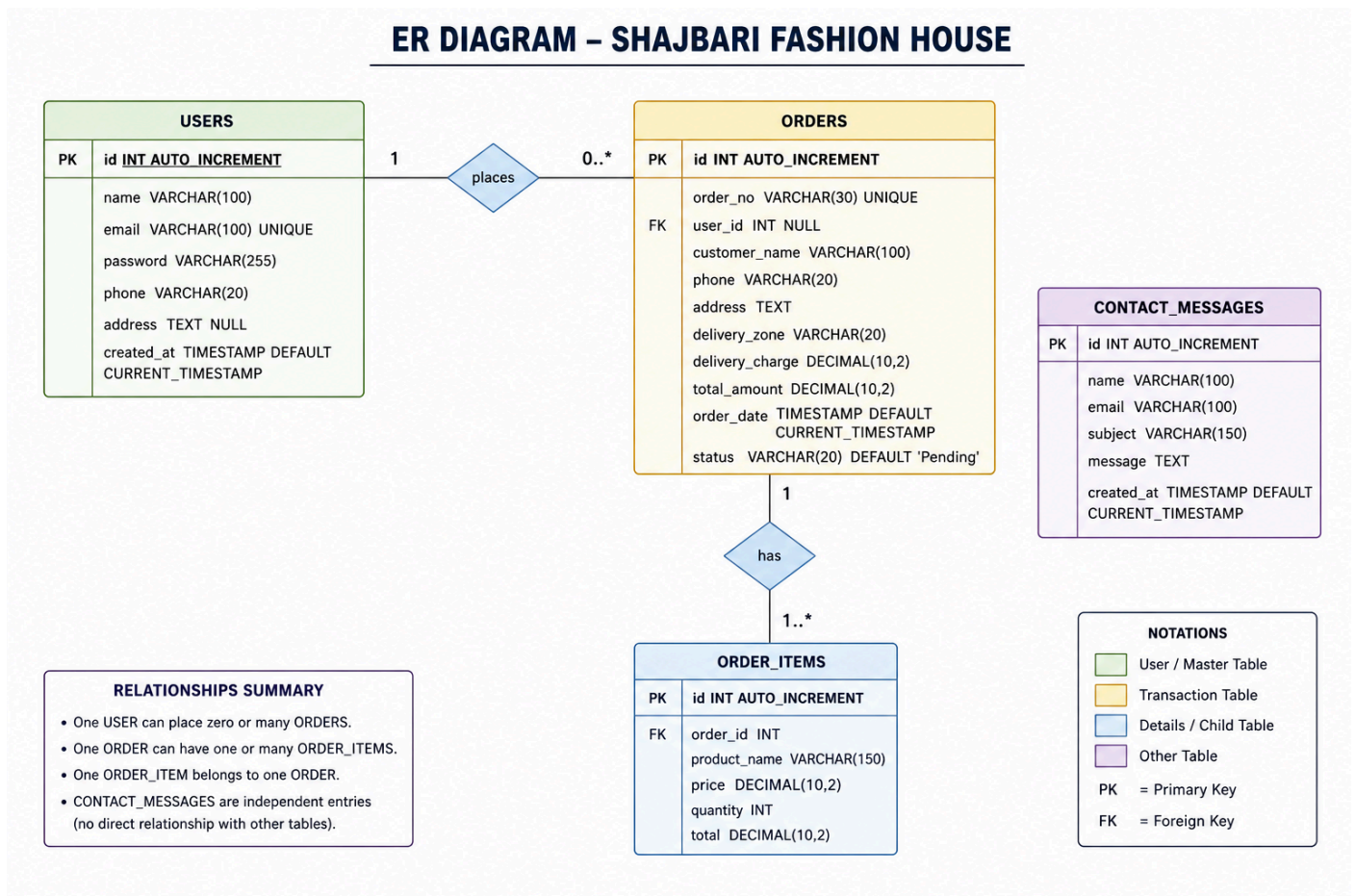
contact_messages

Stores customer contact messages

Relationships

- `orders.user_id` → `users.id`
- `order_items.order_id` → `orders.id`

ER Diagram



Database Schema Explanation

users Table

Stores registered user information including:

- Name
- Email
- Password
- Phone Number
- Id
- created_at

SELECT * FROM `users`

☐ Profiling

[\[Edit inline \]](#)

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[\[Explain SQL \]](#)

[\[Create PHP code \]](#)

[\[Refresh \]](#)

☐ Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

←

→

▼

id

fullname

email

phone

password

created_at

☐

Edit

Copy

Delete

1

Jobaidul Islam

jobaidul@gmail.com

01882643313

\$2y\$10\$beFEzwaC9yY1DTiD.xlvesU7mq/oSM0XFLW4Q6lXD3...

2026-05-09 20:19:31

Edit

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Delete

2

Musfiqur

musfiqur@gmail.com

01700000000

\$2y\$10\$UaulSJn8Y0VNIulobV8gKuCd9lpdBduAqw22rN6CK.J...

2026-05-09 20:53:36

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3

Asif Ahmed

asif@gmail.com

01987654433

\$2y\$10\$Qyp3WgyudYDqq1rzi0nmJuHrPjnfhAQBENOHirGwURu...

2026-05-10 00:17:06

orders Table

Stores:

- Id
- User_id
- customer_name
- Order number
- Delivery charge
- Order status
- Phone
- Customer address
- Subtotal
- Total
- Payment method
- Created_at

`SELECT * FROM `orders``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				id	user_id	order_number	customer_name	phone	address	subtotal	delivery_charge	total	payment_method	status	created_at
☐	Edit	Copy	Delete	1	1	RD260509788	Jobaidul Islam	01882643313	Mohakhali, square building 113/2	24800.00	60.00	24860.00	Cash on Delivery	Shipped	2026-05-09 20:23:37
☐	Edit	Copy	Delete	2	1	RD260509722	musfiq	01529703486	Mirpur 11, dhaka	8000.00	60.00	8060.00	Cash on Delivery	Pending	2026-05-09 22:44:49
☐	Edit	Copy	Delete	3	1	RD260509209	Jobaidul Islam	01529703486	Badda, Dhaka	11800.00	60.00	11860.00	Cash on Delivery	Pending	2026-05-09 22:51:42
☐	Edit	Copy	Delete	4	3	RD260509614	Asif Ahmed	01529703486	Mirpur, shatkhira	21700.00	120.00	21820.00	Cash on Delivery	Delivered	2026-05-10 00:21:23

[↑](#) ☐ Check all | With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

order_items Table

Stores product information related to each order.

Stores:

- Id
- Order_id
- Product_id
- Product_name

- Product_img
- Price
- Quantity

`SELECT * FROM `order_items``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				id	order_id	product_id	product_name	product_img	price	quantity
☐				1	1	1	Royal Jamdani Saree		3500.00	1
☐				2	1	2	Pink Blossom Jamdani		4200.00	2
☐				3	1	3	Golden Silk Saree		6500.00	1
☐				4	1	7	Eid Special Embroidered Panjabi		3200.00	2
☐				5	2	1	Royal Jamdani Saree	jamdani1.jpg	3500.00	1
☐				6	2	8	Luxury Embroidered Panjabi	embroydary2.jpg	4500.00	1
☐				7	3	5	Classic White Cotton Panjabi	cotton1.jpg	1800.00	1
☐				8	3	12	Green Benaroshi Saree	benaroshi4.jpg	6200.00	1
☐				9	3	15	Wedding Embroidered Panjabi	embroydary3.jpg	3800.00	1
☐				10	4	1	Royal Jamdani Saree	jamdani1.jpg	3500.00	1
☐				11	4	2	Pink Blossom Jamdani	jamdani2.jpg	4200.00	1
☐				12	4	7	Eid Special Embroidered Panjabi	embroydary1.jpg	3200.00	1
☐				13	4	11	Crimson Benaroshi Saree	benaroshi3.jpg	7000.00	1
☐				14	4	13	Navy Blue Cotton Panjabi	cotton3.jpg	1900.00	2

contact_messages Table

Stores customer inquiries and messages.

`SELECT * FROM `contact_messages``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				id	name	email	message	is_read	created_at
☐				1	Jobaidul Islam	mdjobaidulislamsabbir2022@gmail.com	Hi my name is jobaidul islam	0	2026-05-09 20:21:46
☐				2	asif	asif@gmail.com	Hi i am asif. your website is good and also your p...	0	2026-05-10 00:19:56

Normalization

The database design follows normalization principles to reduce data redundancy and improve data consistency.

7. Technologies Used

Technology	Purpose
HTML5	Structure of webpages
CSS3	Styling and responsiveness
JavaScript	Client-side interactivity
PHP	Backend development
MySQL	Database management
phpMyAdmin	Database administration
XAMPP	Local server environment

8. Features of the System

Frontend Features

- Sticky navigation bar
- Hero section with statistics
- Product category filters
- Live product search
- Product modal popup
- Shopping cart sidebar
- Quantity management
- Delivery charge selection
- Responsive design
- AJAX contact form
- Toast notifications

Backend Features

- Secure user registration
- Login authentication
- Session management
- Order processing
- JSON API endpoints
- Order history retrieval
- Database-driven functionality

9. Security Practices

Security Practice	Purpose
<code>password_hash()</code>	Encrypt passwords
<code>password_verify()</code>	Verify passwords securely
Prepared statements	Prevent SQL injection
Session management	Maintain login state
Input trimming	Remove unnecessary spaces
Email validation	Validate email addresses
Type casting	Prevent invalid data input

The project follows secure coding practices to protect user data and maintain application security.

10. User Interface and Implementation Details

User Registration System

The registration system allows users to create accounts securely using hashed passwords.

Fashion Store

Create your account and start shopping stylish Saree & Panjabi collections.

Create Account

Full Name

Email Address

Phone Number

Password

Confirm Password

Register

Already have an account? [Login](#)

Login Authentication

The login system verifies user credentials using PHP sessions and password verification methods.

Login

Email

Password

Login

Don't have an account? [Register](#)

Shopping Cart System

Users can add products to the cart, modify quantities, and place orders.

OUR COLLECTIONS

Explore Products

All Products

Jamdani Saree

Silk Saree

Cotton Panjabi

Embroidered Panjabi

Bestseller



JAMDANI SAREE
Royal Jamdani Saree

₹3,500 ~~₹4,200~~ 17% off

Details

Add to Cart

New



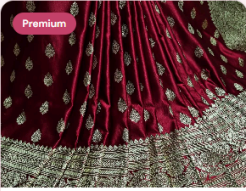
JAMDANI SAREE
Pink Blossom Jamdani

₹4,200 ~~₹5,000~~ 16% off

Details

Add to Cart

Premium



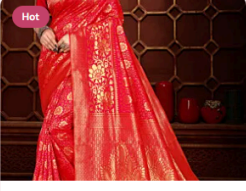
SILK SAREE
Golden Silk Saree

₹6,500 ~~₹8,000~~ 19% off

Details

Add to Cart

Hot



SILK SAREE
Royal Blue Silk Saree

₹5,800 ~~₹7,000~~ 17% off

Details

Add to Cart

Bestseller



New



Eid Special



Premium



Order Management

Orders are stored in the database along with customer and delivery information.

 **Ranga Dhaga**
Fashion House

← Back to Shop

Logout

J

Jobaidul Islam
jobaidul@gmail.com

3 Orders

₹45k Spent

0 Delivered

Phone

01882643313

Member Since

May 9, 2026

Profile Info

Order History

Order History

3 orders

RD260509209

May 9, 2026

Pending

Pending

Confirmed

Processing

Shipped

Delivered



Classic White Cotton Panjabi

× 1 ₹1,800



Green Benaroshi Saree

× 1 ₹6,200



Wedding Embroidered Panjabi

× 1 ₹3,800

Badda, Dhaka

01529703486

Cash on Delivery

Subtotal ₹11,800 + Delivery ₹60

₹11,860

Contact System

The contact form sends user messages to the database without reloading the page.

GET IN TOUCH

Contact Us

 **Phone**
+880 1882-643313

 **Email**
info@shajbari.com

 **Address**
Banani, Dhaka, Bangladesh

Your Name

Your Email

Your Message

Send Message

Bringing Bangladesh's finest textile heritage to your doorstep since 2024.

Quick Links

Home
Products
About Us
Contact

Categories

Jamdani Saree
Silk Saree
Cotton Panjabi
Embroidered Panjabi

Policy

Return Policy
Privacy Policy
Delivery Info
FAQ

Profile System

← Back to Shop

Logout

Profile Info

Order History

Personal Information

FULL NAME

Jobaidul Islam

EMAIL ADDRESS

jobaidul@gmail.com

PHONE NUMBER

01882643313

MEMBER SINCE

May 9, 2026

Account Security

PASSWORD

ACCOUNT STATUS

✓ Active



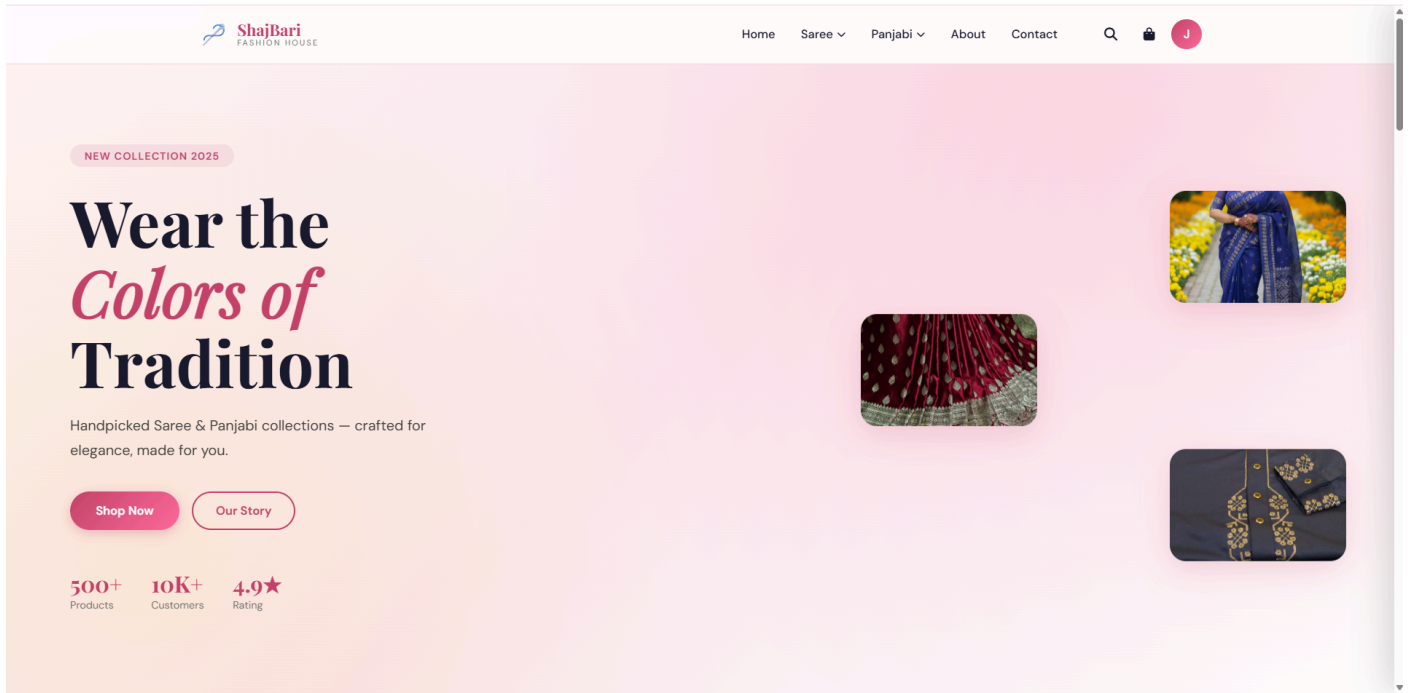
Jobaidul Islam
jobaidul@gmail.com

3 Orders
৳45k Spent
0 Delivered

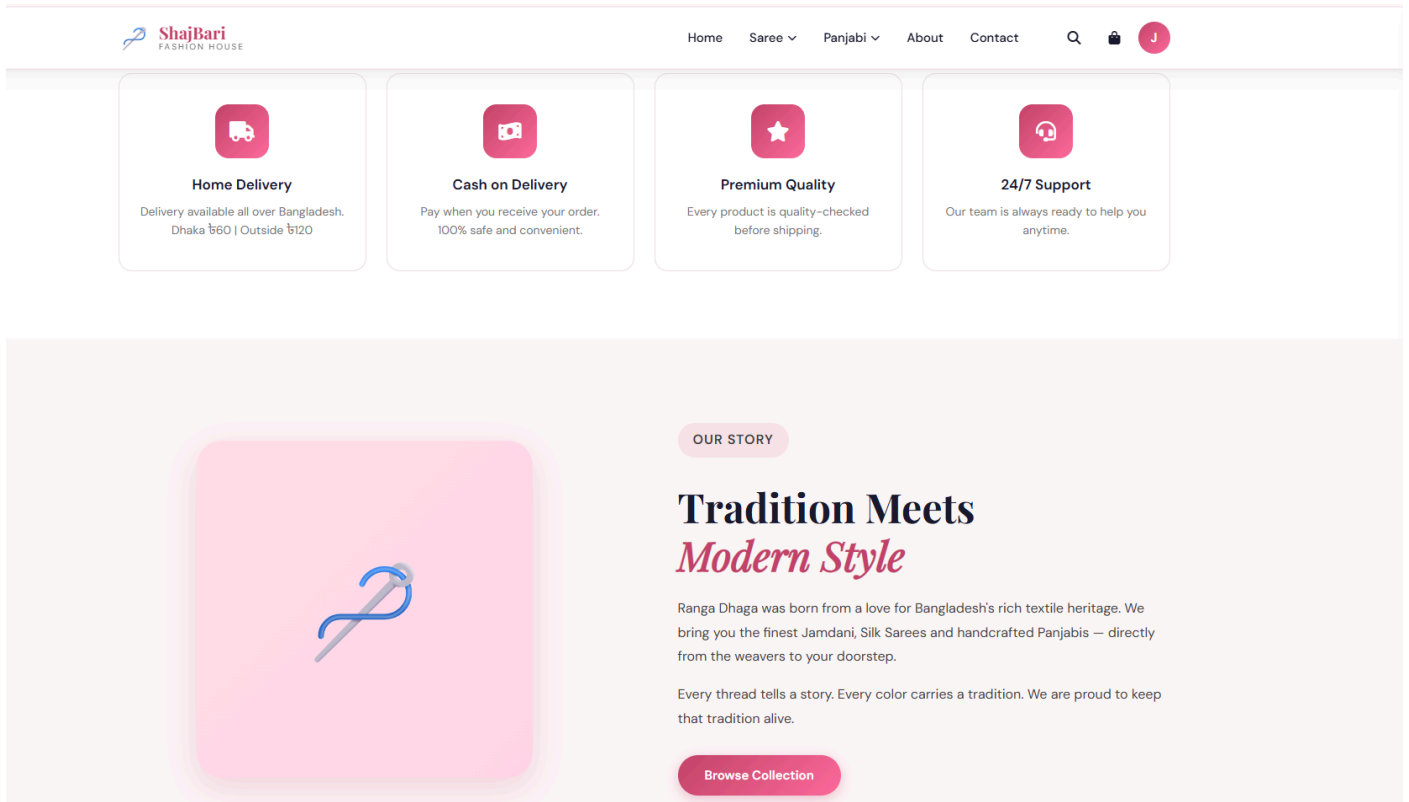
 Phone
01882643313

 Member Since
May 9, 2026

Home



About



11. Testing and Validation

The project was tested to ensure proper functionality.

Test Case	Expected Result	Status
User Registration	Account created successfully	Passed
User Login	Successful authentication	Passed
Add to Cart	Product added correctly	Passed
Place Order	Order stored in database	Passed
Contact Form	Message saved successfully	Passed
Session Management	User remains logged in	Passed

12. Challenges Faced

Several challenges were faced during the development process:

- Managing PHP sessions correctly
- Implementing AJAX communication
- Connecting frontend and backend systems
- Designing a responsive interface
- Maintaining database relationships
- Preventing SQL injection attacks

These challenges helped improve problem-solving and development skills.

13. Future Improvements

Security Enhancements

- CSRF protection
- HTTPS implementation
- Rate limiting
- Advanced validation

User Features

- Password reset system
- Email verification
- Wishlist functionality
- Profile editing

Shopping Features

- Product review system
- Coupon and discount support
- Stock management
- Dynamic product database

Payment Integration

- bKash integration
- Nagad integration
- SSLCommerz payment gateway

Admin Panel

- Product management dashboard
- Order management system
- Revenue tracking
- Customer management

14. Conclusion

ShajBari Fashion House is a successful implementation of a full-stack e-commerce web application developed for the Database Management System course project. The system demonstrates practical use of database management concepts, authentication systems, relational databases, and modern web technologies.

The project provided valuable hands-on experience in frontend development, backend programming, and da

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13. [FreeCodeCamp Web Development Tutorials](#)
14. Database System Concepts — Abraham Silberschatz
15. Modern Web Development practices and online documentation resources were followed during the implementation of this project with Calude and ChatGpt AI.